

Krill Oil for Knee Osteoarthritis

A Randomized Clinical Trial

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IMPORTANCE Knee osteoarthritis is disabling, with few effective treatments. Preliminary evidence suggested that krill oil supplementation improved knee pain, but effects on knee osteoarthritis remain unclear.

OBJECTIVE To evaluate efficacy of krill oil supplementation, compared with placebo, on knee pain in people with knee osteoarthritis who have significant knee pain and effusion-synovitis.

DESIGN, SETTING, AND PARTICIPANTS Multicenter, randomized, double-blind, placebo-controlled clinical trial in 5 Australian cities. Participants with clinical knee osteoarthritis, significant knee pain, and effusion-synovitis on magnetic resonance imaging were enrolled from December 2016 to June 2019; final follow-up occurred on February 7, 2020.

INTERVENTIONS Participants were randomized to 2 g/d of krill oil (n = 130) or matching placebo (n = 132) for 24 weeks.

MAIN OUTCOMES AND MEASURES The primary outcome was change in knee pain as assessed by visual analog scale (range, 0-100; 0 indicating least pain; minimum clinically important improvement = 15) over 24 weeks.

RESULTS Of 262 participants randomized (mean age, 61.6 [SD, 9.6] years; 53% women), 222 (85%) completed the trial. Krill oil did not improve knee pain compared with placebo (mean change in VAS score, -19.9 [krill oil] vs -20.2 [placebo]; between-group mean difference, -0.3; 95% CI, -6.9 to 6.4) over 24 weeks. One or more adverse events was reported by 51% in the krill oil group (67/130) and by 54% in the placebo group (71/132). The most common adverse events were musculoskeletal and connective tissue disorders, which occurred 32 times in the krill oil group and 42 times in the placebo group, including knee pain (n = 10 with krill oil; n = 9 with placebo), lower extremity pain (n = 1 with krill oil; n = 5 with placebo), and hip pain (n = 3 with krill oil; n = 2 with placebo).

CONCLUSIONS AND RELEVANCE Among people with knee osteoarthritis who have significant knee pain and effusion-synovitis on magnetic resonance imaging, 2 g/d of daily krill oil supplementation did not improve knee pain over 24 weeks compared with placebo. These findings do not support krill oil for treating knee pain in this population.

TRIAL REGISTRATION Australian New Zealand Clinical Trials Registry Identifier: ACTRN12616000726459; Universal Trial Number: U1111-1181-7087

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Knee osteoarthritis (OA) affects approximately 654 million people aged 40 years or older worldwide and is associated with pain, functional limitations, and disability.¹ No medical therapies improve the natural history of OA.

Reducing inflammation may effectively treat OA.² Excess synovial fluid within the joint space (effusion) and/or thickening of the synovium (synovitis) are associated with greater pain and structural abnormalities, including cartilage defects, cartilage loss, bone marrow lesions, and incident knee OA.³⁻⁵ Because effusion-synovitis is modifiable, reducing inflammation may slow deleterious changes in structures to improve outcomes in OA.

Consuming marine omega-3 polyunsaturated fatty acids like fish oil, which is high in eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), reduces inflammation.⁶ In people with rheumatoid arthritis, fish oil reduced joint pain, morning stiffness, number of painful/tender joints, and nonsteroidal anti-inflammatory drug (NSAID) use.⁷ However, studies of these therapies in OA have yielded conflicting results.^{8,9} Another source of marine omega-3 fatty acid is Antarctic krill. Both krill oil and fish oil contain omega-3 fatty acids; however, krill oil has better bioavailability than fish oil^{10,11} and contains the antioxidant astaxanthin, which may reduce free radicals, free oxygen, and inflammation.^{11,12} Thus, the combination of omega-3 fatty acids and astaxanthin in krill oil may be a better treatment for OA than fish oil.

In 3 randomized clinical trials (RCTs), daily krill oil reduced some aspects of knee pain, functional impairment, and stiffness among people with knee OA ($n = 56$ and $n = 235$) and in those with mild knee pain ($n = 47$).¹³⁻¹⁵ None of these clinical trials measured effusion-synovitis, a surrogate measure for structural progression of OA.

This RCT evaluated the effect of 2 g/d of krill oil vs identical placebo on the primary outcome of knee pain over 24 weeks in participants with knee OA who had significant knee pain and effusion-synovitis.

Methods

Design, Setting, and Participants

The KARAOKE trial was a multicenter, placebo-controlled, parallel-group, double-blind RCT conducted over 24 weeks. The trial protocol and statistical analysis plan are available in [Supplement 1](#) and [Supplement 2](#), respectively. Details of the trial design were published previously.¹⁶ The trial protocol was approved by the Health and Medical Human Ethics Research Committee (H0015465). Ethics approval was obtained at each site.¹⁶ All participants provided written informed consent. The CONSORT reporting guideline for RCTs was followed.

Participants were recruited from December 2016 to June 2019 via the OA Clinical Trial Network at Australian public hospitals in Melbourne, Sydney, Adelaide, and Perth and at a research institute in Hobart. Recruitment methods involved collaboration with general practitioners, rheumatologists, and orthopedic surgeons as well as advertising through local and social media. Final follow-up occurred on February 7, 2020.

Key Points

Question In people with knee osteoarthritis who have significant knee pain and effusion-synovitis present on magnetic resonance imaging, does krill oil supplementation, compared with placebo, improve knee pain?

Findings In this randomized clinical trial that included 262 adults, compared with placebo, 2 g/d of krill oil supplementation did not improve knee pain as measured on a visual analog scale (range, 0-100, with 0 indicating least pain) (mean difference, 0.30; 95% CI, -6.9 to 6.4; $P = .94$) over 24 weeks.

Meaning This study does not support 2 g/d of krill oil supplementation for improving knee pain in people with knee osteoarthritis who have significant knee pain and effusion-synovitis.

Participants were eligible if they were aged 40 years or older, had symptomatic knee OA (according to American College of Rheumatology criteria),¹⁷ reported significant knee pain (defined as a self-reported pain score ≥ 40 on a visual analog scale [VAS]; range, 0-100, with 100 indicating worst pain) at the screening visit, and had any effusion-synovitis present on magnetic resonance imaging (MRI) (grade ≥ 1 using a modified Whole-Organ Magnetic Resonance Imaging Score; range, 0-3, with 3 indicating severe abnormality).^{4,18} The index knee was the knee with symptomatic OA meeting all inclusion criteria. If both knees met these criteria, the knee with the highest pain score was selected by the study physician.

Exclusion criteria were history of significant knee injury (eg, arthroscopy or open surgery in the index knee < 12 months preceding the study); grade 3 radiographic changes (Altman atlas);¹⁹ other forms of inflammatory arthritis; allergy to seafood; unwillingness to stop taking krill and/or fish oil medications 30 days prior to and during the trial (minimum washout of 4 weeks); use of anticoagulants, high-dose aspirin, or NSAIDs; receipt of a corticosteroid injection in the last 3 months; receipt of a hyaluronic acid injection in the study knee in the last 6 months; contraindication to MRI; pregnancy or breastfeeding; and inability to provide written informed consent.¹⁶

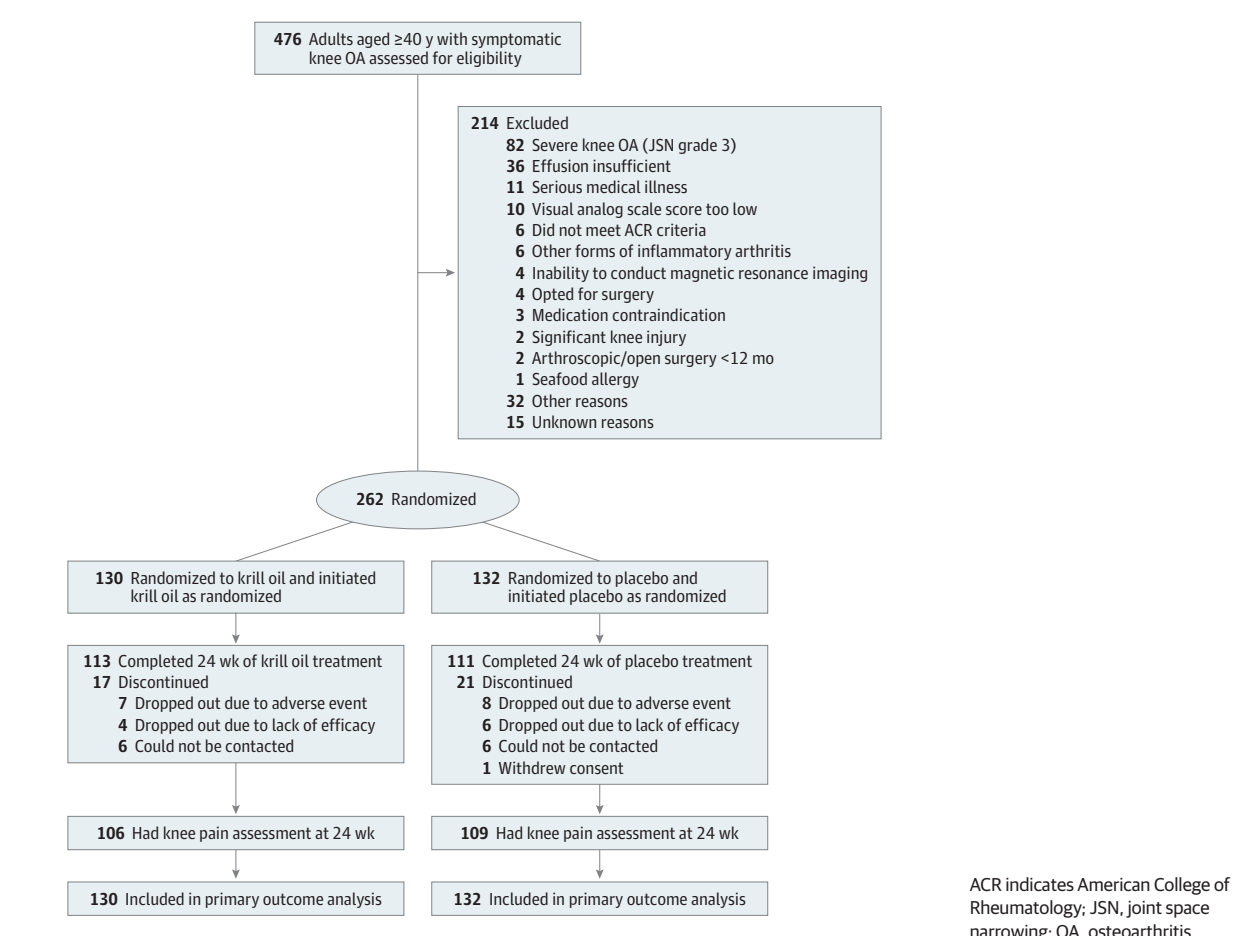
Randomization

Participants were randomized to krill oil or placebo (**Figure 1**) in a 1:1 ratio using a computer-generated random numbers list with adaptive allocation (minimization)²⁰ independently of the investigators. Randomization was stratified by study site. Treatment assignment was masked for participants, research coordinators, and investigators, and masking was maintained until data collection was completed.¹⁶

Interventions

The treatment group received two 1-g softgels daily of krill oil, each softgel containing 190 mg/g of EPA and 100 mg/g of DHA, supplying a total of 350 mg/g of omega-3 content and 12 mg/g of omega-6 content (ratio of 1:29). The placebo group received a combination of vegetable oils (virgin cold-pressed olive oil, maize oil, palm kernel oil, and medium chain triglycerides) containing no EPA or DHA and less than 5 mg/g (0.5%)

Figure 1. Participant Flow in the KARAOKE Trial



of other omega-3 content.¹⁶ Both krill oil and placebo were administered in nondistinguishable glycerin softgels.

Study Assessments

Following the screening visit, study visits occurred at weeks 0 (baseline), 12, and 24.¹⁶ MRI scans were obtained during screening and at 24 weeks, knee radiography was performed at screening, and blood samples were taken at screening (mean gap between screening and week 0, 52 days), 12 weeks, and 24 weeks. Additionally, questionnaires were mailed at 4, 8, 16, and 20 weeks.

Primary Outcome

The primary outcome was 24-week change in self-reported knee pain score measured on a VAS over the past 7 days (range, 0-100, with 0 indicating no pain and 100 indicating the worst pain; minimum clinically important improvement [MCII] = 15).²¹

Secondary Outcomes

Secondary outcomes included change in effusion-synovitis volume on MRI at 24-week follow-up (MCII unknown); change in study knee pain at 4-, 8-, 12-, 16-, and 20-week follow-up; change in hand and back pain VAS scores at all time points

(range, 0-100²²; MCII = 15)²¹; change in Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) scores for knee pain (total pain, weight-bearing pain, and non-weight-bearing pain measured separately; score range, 0-500, with 500 indicating worst pain) over the past 7 days; and change in WOMAC function scores (range, 0-1700, with 1700 indicating worst pain) over the past 7 days at all follow-up time points (MCII = 12% increase from baseline value for each score).^{16,23,24}

An additional secondary outcome was the proportion of participants with response to treatment according to Outcome Measures in Rheumatology-Osteoarthritis Research Society International (OMERACT-OARSI) criteria,²⁵ wherein response was defined as either (1) 50% or greater improvement and an absolute change of 20 points or more in VAS pain score or WOMAC function score or (2) improvement of 20% or greater with an absolute change of 10 mm or more in at least 2 of the following 3 measures at all time points (4, 8, 12, 16, 20, and 24 weeks; MCII unknown): (1) pain; (2) function; and (3) patient global assessment.

Blood markers were assessed at 12 and 24 weeks and included fasting blood glucose²⁶; lipids (high-density lipoprotein cholesterol [MCII = 0.26 mmol/L] and low-density lipoprotein cholesterol [minimum clinically important difference {MCID} = 0.10 mmol/L])²⁶; triglycerides (MCID = 0.09)²⁶; and high-sensitivity C-reactive protein (MCID = 0.5 mg/dL).²⁶

Lower limb strength was assessed as the mean of 2 measurements in both legs simultaneously using a back, leg, and chest dynamometer (TTM Muscular Meter; Tsutsumi Co Ltd) (MCID unknown) at 12 and 24 weeks.

Additional secondary outcomes were a quality-of-life measure, the Assessment of Quality of Life 6D score, at all time points (range, -0.04 to 1.0; MCII = 0.06)²⁷; change in analgesic use at all time points; and adverse events at 12 and 24 weeks.

Secondary outcomes not reported herein due to a large amount of missing data were pressure pain threshold testing and ultrasound-assessed effusion volume. Bone marrow lesion size was a prespecified secondary outcome that was not assessed due to budget constraints; health economic data were a prespecified secondary outcome not assessed because the trial was negative.¹⁶

Post Hoc Outcomes

The Intermittent and Constant Osteoarthritis Pain (ICOAP) score was used to evaluate pain experienced over the past 7 days.²⁸ The score range is 0 to 4, with 0 indicating “never” and 4 indicating “very often.” The ICOAP total pain (range, 0-44), constant pain (range, 0-20), and intermittent pain (range, 0-24) scores were all rescaled to a range of 0 to 100, with 100 indicating worst pain. The MCII are 18.5, 18.7, and 18.4, respectively, using a patient global assessment of change of “somewhat better.”²⁹

MRI Assessments

MRI scans of the index knee were performed at baseline and 24-week follow-up using a 1.5T or 3T whole-body magnetic resonance unit and a dedicated knee coil. Sequences and parameters of MRI units at each site were described previously.¹⁶

Knee effusion-synovitis was defined as presence of an intra-articular fluid-equivalent signal on T2-weighted MRI sequences in any compartment of the knee visible on MRI.¹⁶ Effusion-synovitis volume was measured in milliliters using a semiautomated segmentation method according to the intra-articular fluid-equivalent signal on a section-by-section basis in each region of interest by a coinvestigator (G.C.) (eFigure in Supplement 3). The intraclass correlation for effusion-synovitis was 0.96 (95% CI, 0.94-0.97) among 50 randomly selected participants (range, 0-1, with 1 indicating perfect agreement).

Three-dimensional volume rendering was generated using commercial in-house imaging software (Mimics version 21.0). A modified Whole-Organ Magnetic Resonance Imaging Score was used to assess effusion-synovitis (grade 0-3) semiquantitatively in 4 regions of interest (suprapatellar pouch, central portion, posterior femoral recess, and subpopliteal recess) in terms of the estimated maximum distention of the synovial cavity. The highest score for any of the 4 regions of interest was used to define the maximum effusion of the knee.

Data on concomitant medications were obtained by questionnaire at 12 and 24 weeks. Participants were asked to continue using concomitant medications (including NSAIDs) without changing dose and to use paracetamol (acetaminophen) if they experienced discomfort requiring medication.

Omega-3 Blood Levels and Study Drug Adherence

Omega-3 is a marker of adherence to krill oil and was analyzed from stored blood according to the protocol of Harris and Polreis.³⁰ The Omega-3 Index was calculated by measuring percentages of EPA and DHA in red blood cell membranes as a proportion of total fatty acids. An Omega-3 Index of 8% or greater is associated with lower cardiovascular risk. Treatment adherence was also assessed by counting remaining pills in returned bottles at 12- and 24-week follow-up.^{16,31}

Adverse Events

Data on adverse events were obtained at baseline and at each study visit until the final visit or study withdrawal. Adverse events were defined as any untoward event occurring during the trial regardless of whether it was considered treatment related. Serious adverse events were defined as death; life-threatening, disabling, nonelective, or prolonged hospitalization; and important medical events such as a new cancer diagnosis. Participants were asked to report any adverse event to the research staff during each visit and spontaneously if necessary. New conditions or medical procedures were reported as adverse events. All reported adverse and serious adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA; <https://www.meddra.org>). Participants who were randomized and received at least 1 dose of either krill oil or placebo were included in outcome analyses.

Other Assessments

Radiographic knee OA was assessed at screening using radiography per the Osteoarthritis Research Society International atlas.¹⁹ Height and weight were measured and body mass index was calculated as weight in kilograms divided by height in meters squared.¹⁶

Power Calculations

Based on data from other clinical trials,^{32,33} we estimated that a sample size of 130 per group provided 90% power to detect a 10-point difference on the VAS (SD for pain change, 23.8 points) and a 4.5-mL change in effusion-synovitis volume between krill oil and placebo, at a 5% probability of type I error ($\alpha = .05$).

Statistical Analyses

The primary and secondary outcomes were evaluated by applying a linear mixed model with treatment, month, and their interaction (treatment $\times$ month) as covariates, as well as outcomes measured at baseline if there were differences at baseline (Table 1). This model included all trial participants who underwent randomization and had at least 1 value for the corresponding outcome. To address effects of missing data, we assessed variables related to missingness. In the primary outcome analyses, age was the only variable associated with missingness. Therefore, as a sensitivity analysis, we adjusted for the effect of age and missingness induced by age. No meaningful difference in the results was observed (eTable 1 in Supplement 3). Individual participant and trial center were treated as random intercepts and month was treated as a random slope

Table 1. Baseline Participant Characteristics

Characteristics	Krill oil (n = 130)	Placebo (n = 132)
Age, mean (SD), y	61.4 (9.9)	61.7 (9.3)
Female sex, No. (%)	64 (49)	76 (58)
Male sex, No. (%)	66 (51)	56 (42)
Body mass index, mean (SD) ^a	29.3 (5.6)	29.9 (6.5)
≥35, No (%)	16 (13)	25 (20)
Knee pain VAS score, mean (SD) ^b	47.8 (20.3)	50.2 (20.4)
Effusion-synovitis volume, median (IQR), mL	4.3 (2.5-9.2)	5.2 (2.8-11.0)
Whole-Organ Magnetic Resonance Imaging Score, No. (%) ^c		
1	42 (33)	38 (29)
2	29 (23)	25 (19)
3	57 (45)	66 (51)
WOMAC score, mean (SD) ^d		
Pain	186.3 (98.2)	209.6 (99.6)
Function	575.7 (312.4)	617.9 (333.2)
Hand pain VAS score, mean (SD) ^b	21.7 (26.3)	24.4 (28.5)
Back pain VAS score, mean (SD) ^b	26.7 (25.9)	33.0 (30.5)
Lower limb strength, mean (SD), N ^e	66.5 (38.1)	65.8 (37.1)
AQoL-6D score, mean (SD) ^f	0.77 (0.15)	0.75 (0.17)
ICOAP score, mean (SD) ^g		
Intermittent	39.9 (18.4)	40.4 (20.2)
Constant	27.7 (20.3)	32.7 (22.1)
Total	34.3 (17.2)	36.7 (19.3)
Pressure pain threshold of the study knee, median (IQR), kPa/s ^h	4.2 (3.1-6.1)	4.3 (3-6.2)
Medication use, No. (%) ⁱ		
Acetaminophen	38 (30)	35 (27)
Nonsteroidal anti-inflammatory drugs	25 (20)	40 (31)
Glucosamine	11 (9)	19 (15)
Turmeric	5 (4)	8 (6)
Laboratory measures ^j		
High-sensitivity C-reactive protein, median (IQR), mg/dL	1.4 (0.7-2.9)	1.9 (1.0-3.8)
Triglycerides, mean (SD), mmol/L	1.4 (1.1)	1.3 (0.7)
HDL cholesterol, mean (SD), mmol/L	1.6 (0.4)	1.5 (0.4)
LDL cholesterol, mean (SD), mmol/L	3.0 (0.9)	3.2 (0.9)
Fasting glucose, mean (SD), mmol/L	5.3 (0.7)	5.3 (1.0)
Omega-3 Index, mean (SD), % ^k	6.5 (1.6)	6.5 (1.5)
Study site, No. (%)		
Hobart	64 (49.2)	61 (46.2)
Melbourne	27 (20.8)	27 (20.5)
Adelaide	11 (8.5)	19 (14.4)
Sydney	6 (4.6)	4 (3.0)
Perth	22 (16.9)	21 (15.9)

Abbreviation: VAS, visual analog scale.

SI conversion factors: To convert triglycerides to mg/dL, divide by 0.0113; high (HDL)- and low-density lipoprotein (LDL) cholesterol to mg/dL, divide by 0.0259; glucose to mg/dL, divide by 0.0555.

^a Calculated as weight in kilograms divided by height in meters squared.

^b Self-reported VAS pain score over the past 7 days (range, 0 [no pain]-100 [unbearable pain]; minimum clinically important difference [MCID] = 15).

^c Score range, 0 (no abnormality) to 3 (severe abnormality).

^d The Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) is self-administered with 5 questions on pain and 17 questions on physical function; scale range for each question, 0 (none) to 100 (unbearable). Composite scores for pain: 0-500 (MCID = 12% increase of baseline value); for function, 0-1700 (MCID = 12% increase in baseline value).²³

^e Assessed in both legs simultaneously using a dynamometer. It is done in duplicate, with the mean of the 2 estimates used. A higher value indicates a greater force (N) exerted, ie, better leg strength. The MCID is unknown.

^f The Assessment of Quality of Life (AQoL-6D) is self-administered with 20 quality-of-life questions³⁴ to create a utility score that ranges from 1.00 (full health) to 0.00 (death-equivalent health states) to -0.04 (health states worse

than death). The MCID is 0.06.

^g The Intermittent and Constant Osteoarthritis Pain (ICOAP) is self-administered with 5 questions on constant knee pain and 8 on knee pain that comes and goes on average in the past week.²⁸ Question responses are "not at all," "mildly," "moderately," "severely," and "extremely." Composite scores range from 0 (no pain)-100 (unbearable pain).

^h Pressure pain threshold determined by the mean of the threshold from a series of 3 stimulus intensities, each applied as a slowly increasing ramp of 50 kPa/s on the study knee. Testing was conducted only on the study knee at the Melbourne and Perth sites (krill oil: n = 43; placebo: n = 40).

ⁱ Defined as taking any regular medication or dietary supplements.

^j Reference values: high-sensitivity C-reactive protein, <5 mg/dL; fasting glucose, 5.6 mmol/L. Target values: triglycerides, <2.0 mmol/L; HDL, ≥1.0 mmol/L; LDL, <2.0 mmol/L.

^k Omega-3 Index calculated as the sum of percentages of eicosapentaenoic acid and docosahexaenoic acid in the total fatty acids present in red blood cell membranes, expressed as a percentage of total fatty acids. The target Omega-3 Index for optimal health is under debate, but an index of 8% is suggested to be associated with lower cardiovascular disease risk.

Table 2. Change in Study End Points Over 24 Weeks^a

Outcomes	Krill oil, mean (95% CI) (n = 130)		Placebo, mean (95% CI) (n = 132)		Between-group difference in change, mean (95% CI)	P value		
	Baseline	24 wk	Change	24 wk				
Primary outcome								
Knee pain, VAS score ^b	47.8 (44.3-51.3)	26.5 (22.0-31.1)	-19.9 (-24.7 to -15.2)	29.3 (24.4-34.2)	-20.2 (-24.9 to -15.5)	.94		
Secondary outcomes								
Effusion-synovitis volume, median (IQR), mL ^c	4.3 (2.5-9.2)	5.4 (2.5-10.7) [n = 129]	0.81 (-0.17 to 1.79)	5.7 (3.1-9.7) [n = 128]	-0.94 (-1.92 to 0.04)	.01		
WOMAC score ^d								
Total pain	186 (170-203)	110 (89-130) [n = 129]	-87 (-110 to -63)	210 (192-227)	130 (107-152) [n = 130]	-83 (-106 to -59)	3.0 (-24 to 31)	.81
Weight-bearing pain	127 (116-138)	74 (61-87) [n = 106]	-51 (-61 to -42)	141 (130-151)	86 (72-101) [n = 111]	-48 (-58 to -39)	3 (-10 to 16)	.70
Non-weight-bearing pain	60 (52-68)	36 (27-44) [n = 106]	-23 (-32 to -15)	69 (61-77)	43 (35-52) [n = 111]	-21 (-29 to -13)	2 (-10 to 14)	.70
Function	578 (524-633)	319 (255-383) [n = 129]	-240 (-299 to -182)	618 (560-676)	419 (345-494) [n = 128]	-189 (-247 to -132)	51 (-31 to 133)	.22
Hand pain, VAS score ^b	21.7 (16.9-26.4)	14.4 (10.6-18.1) [n = 118]	-7.6 (-11.9 to -3.3)	24.4 (19.3-29.6)	17.6 (12.8-22.3) [n = 121]	-6.3 (-10.6 to -2.1)	1.3 (-4.8 to 7.3)	.69
Back pain, VAS score ^b	26.7 (22.0-31.4)	18.4 (14.2-22.7) [n = 120]	-7.6 (-12.2 to -3.0)	33.0 (27.5-38.5)	22.8 (18.0-27.5) [n = 121]	-8.9 (-13.5 to -4.4)	-1.3 (-7.9 to 5.2)	.69
Lower limb strength, N ^d	66.5 (59.8-73.2)	72.8 (65.7-79.8) [n = 126]	6.7 (2.6 to 10.7)	65.9 (59.4-72.3)	71.7 (64.3-79.1) [n = 130]	4.4 (0.5 to 8.4)	-2.2 (-7.9 to 3.4)	.44
AQoL-6D score ^d	0.8 (0.7-0.8)	0.8 (0.8-0.9) [n = 125]	0.04 (0.02 to 0.06)	0.7 (0.7-0.8)	0.8 (0.8-0.8) [n = 129]	0.03 (0.01 to 0.05)	-0.01 (-0.04 to 0.01)	.38
OMERACT-OARS response, No. (%) ^e			50 (47) [n = 107]			45 (41) [n = 110]	1.14 (0.84 to 1.55) ^f	.39
Laboratory measures								
High-sensitivity C-reactive protein, median (IQR), mg/dL	1.4 (0.7-2.9)	1.3 (0.8-2.3) [n = 104]	-0.76 (-1.60 to 0.08)	1.9 (1.0-3.8)	1.9 (1.1-3.3) [n = 103]	-0.12 (-0.97 to 0.73)	0.64 (-0.56 to 1.84)	.30
Triglycerides, mmol/L	1.4 (1.2-1.6)	1.3 (1.2-1.4) [n = 128]	-0.08 (-0.21 to 0.05)	1.3 (1.2-1.4)	1.4 (1.3-1.6) [n = 129]	0.07 (-0.06 to 0.20)	0.15 (-0.04 to 0.33)	.11
High-density lipoprotein cholesterol, mmol/L	1.6 (1.5-1.6)	1.6 (1.5-1.6) [n = 127]	0.02 (-0.03 to 0.06)	1.5 (1.5-1.6)	1.5 (1.4-1.6) [n = 127]	-0.01 (-0.06 to 0.03)	-0.03 (-0.09 to 0.03)	.32
Low-density lipoprotein cholesterol, mmol/L	3.0 (2.9-3.2)	3.0 (2.8-3.2) [n = 126]	-0.01 (-0.12 to 0.10)	3.2 (3.0-3.3)	3.2 (3.0-3.4) [n = 127]	0.0005 (-0.11 to 0.11)	0.01 (-0.14 to 0.17)	.90
Fasting glucose, mmol/L	5.3 (5.2-5.5)	5.4 (5.2-5.6) [n = 127]	0.07 (-0.12 to 0.26)	5.3 (5.1-5.5)	5.4 (5.2-5.6) [n = 125]	0.11 (-0.08 to 0.29)	0.04 (-0.23 to 0.30)	.79
Post hoc outcomes								
ICOAP score ^d								
Constant pain	27.7 (24.2-31.2)	16.6 (12.7-20.5)	-10.7 (-14.4 to -6.99)	32.7 (28.9-36.5)	23.0 (19.0-26.9)	-7.8 (-11.4 to -4.11)	2.95 (-2.29 to 8.18)	.27
Intermittent pain	39.9 (36.7-43.1)	29.2 (25.8-32.7)	-9.7 (-13.5 to -6.0)	40.4 (36.9-43.9)	31.6 (27.9-35.4) [n = 131]	-8.2 (-11.9 to -4.5)	1.5 (-3.8 to 6.9)	.62
Total pain	34.3 (31.3-37.3)	23.4 (20.0-26.8)	-9.9 (-13.3 to -6.57)	36.7 (33.4-40.1)	27.4 (23.8-31.1)	-8.2 (-11.9 to -4.5)	2.05 (-2.64 to 6.74)	.39

Abbreviations: VAS, visual analog scale; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.

^a e Tables 4-6 in Supplement 3 provide results for the additional secondary outcomes. Analyses are adjusted for baseline if $P \leq .10$ for differences in Table 1 (high-sensitivity C-reactive protein, back pain, and WOMAC total pain and weight-bearing pain scores). No values were imputed for missing values. To address the effect of missing data, we assessed variables related to missingness. In the primary outcome analyses, age was the only variable associated with missingness; therefore, a sensitivity analysis was conducted adjusting for the effect of age and the missingness induced by age. No significant difference in the results was observed (eTable 1 in Supplement 3).

^b Knee pain, hand pain, and back pain VAS scores range from 0 (no pain) to 100 (unbearable pain).
^c Effusion-synovitis volume ranges from 0 mL to ∞ ; higher values indicate more effusion-synovitis.
^d See Table 1 footnotes for descriptions of scores and measures for the WOMAC, lower limb strength, Assessment of Quality of Life 6D (AQoL-6D), and Intermittent and Constant Osteoarthritis Pain Score (ICOAP).
^e Response to treatment according to Outcome Measures in Rheumatology-Osteoarthritis Research Society International (OMERACT-OARS) criteria was calculated at the final follow-up visit as a measure of change from baseline.²⁵ See the Methods for response criteria.

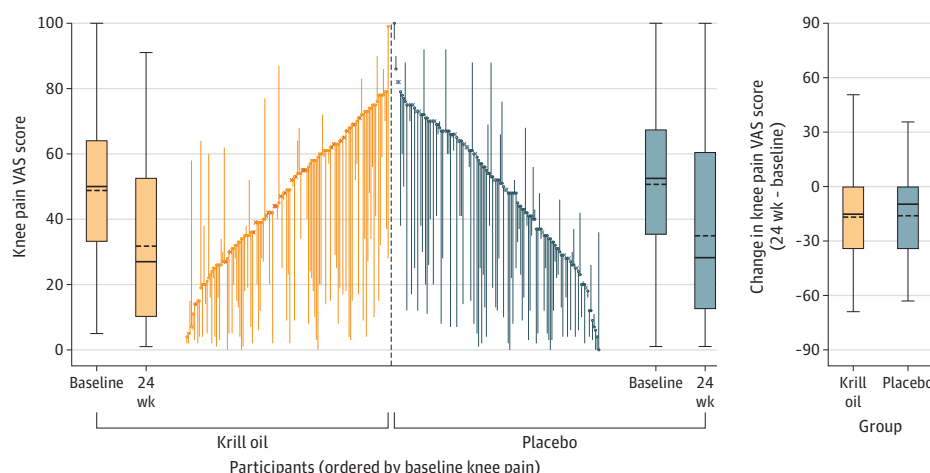
^b Knee pain, hand pain, and back pain VAS scores range from 0 (no pain) to 100 (unbearable pain).

^c Effusion-synovitis volume ranges from 0 mL to ∞ ; higher values indicate more effusion-synovitis.

^d See Table 1 footnotes for descriptions of scores and measures for the WOMAC, lower limb strength, Assessment of Quality of Life 6D (AQoL-6D), and Intermittent and Constant Osteoarthritis Pain Score (ICOAP).

^e Response to treatment according to Outcome Measures in Rheumatology-Osteoarthritis Research Society International (OMERACT-OARS) criteria was calculated at the final follow-up visit as a measure of change from baseline.²⁵ See the Methods for response criteria.

Figure 2. Knee Pain VAS Scores Over 24 Weeks



Participants are arranged in order of baseline knee pain VAS score values, ascending for krill oil and descending for placebo. Lines extend to each participant's 24-week follow-up knee pain VAS score. The X's represent participants with a baseline value but no 24-week follow-up values. Box plots show the distribution of baseline and 24-week knee pain separately for each group. Box plots for each treatment group (right panel) show the distribution of

the calculated change in knee pain from baseline to 24 weeks for each participant (excluding those with missing follow-up data). Boxes are bounded by the first and third quartiles with the median shown as a thicker solid line and the mean shown as a dashed line. Whiskers extend to the most extreme values within 1.5 times the IQR.

to allow for correlated data and different treatment effects among participants over time, respectively. Change in outcome measures within each group and differences in changes between groups from baseline to follow-up were calculated using linear combinations of the estimated coefficients. Post hoc analyses were evaluated parallel to evaluation of other outcomes.

We used Stata version 13.0 (StataCorp) and R version 4.0 (R Foundation) for statistical analyses. A 2-sided $P < .05$ was considered statistically significant. Due to the large number of secondary outcomes, results for secondary outcomes should be considered exploratory.

Results

Of 476 individuals screened, 262 were enrolled; 130 were randomized to receive krill oil and 132 were randomized to receive placebo (Figure 1). Forty participants withdrew or were lost to follow-up (15%), including 17 (13.1%) in the krill oil group and 23 (17.4%) in the placebo group; 222 (84.7%) completed the trial. Numbers of participants who were followed up at each visit were 241 at 4 weeks, 235 at 8 weeks, 230 at 12 weeks, 226 at 16 weeks, 224 at 20 weeks, and 222 at 24 weeks.

The mean age of participants was 61.6 years (range, 40–88 years); 53% were women (Table 1). Groups were comparable in demographic and clinical characteristics (Table 1).

Process Measures

Mean rates of adherence were 96% in the placebo group and 99% in the krill oil group among those who returned softgels; 95% of participants consumed at least 80% of softgels over the 24-week study period (eTable 2 in Supplement 3). The mean

Omega-3 Index increased in the krill oil group from 6.5% at baseline to 7.8% at 12 weeks and 8.0% at 24 weeks; mean levels remained stable in the placebo group at 6.5% at baseline, 6.5% at 12 weeks, and 6.6% at 24 weeks (eTable 3 in Supplement 3).

Primary Outcome

There was no significant difference in change in knee pain VAS scores between the 2 groups from baseline to 24 weeks (−19.9 in the krill oil group vs −20.2 in the placebo group; between-group mean difference, −0.3; 95% CI, −6.9 to 6.4; $P = .94$) (Table 2 and Figure 2).

Secondary and Post Hoc Outcomes

Of the 60 secondary outcomes, compared with krill oil, placebo significantly reduced effusion-synovitis volume at 24 weeks (0.81 mL in the krill oil group vs −0.94 mL in the placebo group; between-group mean difference, −1.75 mL; 95% CI −3.13 to −0.37 mL; $P = .01$). Krill oil significantly improved triglycerides over 12 weeks (−0.11 mmol/L in the krill oil group vs 0.14 mmol/L in the placebo group; between-group mean difference, 0.24 mmol/L; 95% CI, 0.07–0.42 mmol/L; $P = .01$), but not over 24 weeks. There were no significant differences between krill oil and placebo for any other study outcomes (Table 2; eTables 4–6 in Supplement 3).

There were no changes in post hoc outcomes (Table 2).

Adverse Events

Two hundred three adverse events were reported, 97 events among 67 participants in the krill oil group and 106 events among 71 participants in the placebo group. One or more adverse events was reported in 51% of the krill oil group (67/130) and in 54% of the placebo group (71/132) (Table 3).

Table 3. Incidence of Adverse Events and Serious Adverse Events^a

Adverse events	Krill oil, No. (n = 130)	Placebo, No. (n = 132)
Total adverse events	97	106
Musculoskeletal and connective tissue disorders	32	42
Pain in knee	10	9
Pain in extremity	1	6
Pain in hip	3	2
Respiratory, thoracic, and mediastinal	9	16
Upper respiratory tract infection	4	4
Nasopharyngitis	2	1
Cough	2	0
Lower respiratory tract infection	1	1
Gastrointestinal discomfort and disorders	13	6
Gastroesophageal reflux disease	1	3
Abdominal discomfort	0	3
Diarrhea	2	1
Investigations	10	8
Surgical and medical procedures	5	8
Injury, poisoning, and procedural complications	11	6
Skin and subcutaneous tissue disorders	5	3
General system disorders	3	2
Nervous system disorders	2	3
Participants with an adverse event, No. (%) ^b	67 (50.7)	71 (53.8)
Total serious adverse events ^c	9	6
Any treatment-related serious adverse event	0	0

^a Specific categories of adverse events are presented only for events occurring in at least 3 participants. Detailed information is provided in eTable 4 in [Supplement 3](#).

^b No serious adverse events related to treatment were identified by the study investigators.

^c Serious adverse events were defined as death; life-threatening, disabling, nonelective, or prolonged hospitalization; and important medical events such as a new cancer diagnosis. Comprehensive details on serious adverse events can be found in eTables 5 and 6 in [Supplement 3](#).

The most common adverse events were musculoskeletal and connective tissue disorders (32 events [25%] with krill oil and 42 events [32%] with placebo). This included knee pain (10 events with krill oil and 9 events with placebo), lower extremity pain (1 event with krill oil and 5 events with placebo), and hip pain (3 events with krill oil and 2 events with placebo) (Table 3; eTable 7 in [Supplement 3](#)). Other common events include connective tissue disorders and respiratory, thoracic, and mediastinal disorders were most common (eTable 7 in [Supplement 3](#)).

Participants taking krill oil reported more adverse events related to gastrointestinal discomfort and disorders (13 events [10%] with krill oil vs 6 events [4.5%] with placebo). There were 9 serious adverse events in the krill oil group and 6 serious adverse events in the placebo group. All events were considered unrelated to treatment (eTable 8 in [Supplement 3](#)).

Discussion

Supplementation of 2 g/d of krill oil did not significantly improve knee pain over 24 weeks compared with placebo. These data do not support use of 2 g/d of krill oil for relieving knee pain in people with knee OA who have significant knee pain and effusion-synovitis.

The hypothesis that marine-sourced omega-3 polyunsaturated fatty acids EPA and DHA from krill oil may reduce pain and effusion-synovitis in knee OA was based on the role of inflammation in OA pathogenesis and findings that suggested benefit from krill oil in 2 small RCTs.^{13,14} Suppressing inflammation could attenuate adverse effects of inflammation, reduce pain, and slow deleterious changes in knee structure. However, data reported herein demonstrated no effect of krill oil on pain or effusion-synovitis volume, indicating no change in intra-articular inflammation.

These results differ from results of 3 previous RCTs, in which krill oil decreased knee pain.¹³⁻¹⁵ Two small placebo-controlled RCTs^{13,14} suggested that krill oil supplementation for 30 days decreased knee pain. Deutsch¹³ showed that 300 mg/d of krill oil significantly decreased C-reactive protein levels, pain, stiffness, and functional impairment in 90 people with cardiovascular disease, rheumatoid arthritis, and/or OA and with increased levels of C-reactive protein (>1.0 mg/dL) as measured by WOMAC. However, the population included participants with conditions other than OA (rheumatoid arthritis and cardiovascular disease), and the report did not present any subgroup analyses. Suzuki et al¹⁴ reported that intake of 2 g/d of krill oil in 50 people with mild knee pain (not necessarily knee OA) decreased knee pain and stiffness but did not change any blood parameters, including white blood cell count and C-reactive protein.

Stonehouse et al¹⁵ reported that daily krill oil supplementation (4 g/d) was safe and modestly improved knee pain, stiffness, and physical function in 235 adults with mild to moderate knee OA over 26 weeks. Differences in findings by Stonehouse et al¹⁵ compared with the results reported here might be explained by differing EPA and DHA content in the intervention groups. In the current study, participants received 2 g/d of krill oil containing 380 mg of EPA and 200 mg of DHA, which led to an Omega-3 Index value of 8.0%, whereas participants in the study by Stonehouse et al¹⁵ received 4 g/d of krill oil containing 880 mg/d of EPA and DHA (600 mg of EPA and 280 mg of DHA), which resulted in an Omega-3 Index value of 9.0%. An Omega-3 Index of 8.0% or greater may be cardioprotective³⁵; however, a specific Omega-3 Index required to change clinical or structural outcomes in knee OA has not been identified. Stonehouse et al.¹⁵ did not require effusion-synovitis for clinical trial inclusion. Another potential explanation for the different findings reported herein compared with the trial by Stonehouse et al¹⁵ is that the baseline Omega-3 Index in participants in the current study was higher compared with that study (6.5% vs 5.7%). The trial by Stonehouse et al¹⁵ limited dietary intake of omega-3 long-chain polyunsaturated fatty acids to less than 500 mg/d among participants.

In contrast, participants in the current clinical trial were not required to limit dietary omega-3 intake.

In this trial, participants receiving placebo demonstrated significant reductions in effusion-synovitis volume compared with participants receiving krill oil, whereas no differences were observed in pain or in any secondary outcomes between the 2 groups. The reason for this unexpected outcome remains unclear. The comparator oil used consisted of a blend of vegetable oils, which did not contain EPA or DHA and contained less than 5 mg/g (0.05%) of other omega-3 content. The composition was similar to the placebo oil used by Stonehouse et al.¹⁵ Given the large number of study outcomes in this clinical trial, this finding may have been due to chance.

Limitations

This study has several limitations. First, the krill oil dose of 2 g/d may have been insufficient to generate sufficiently large anti-inflammatory effects to reduce pain and volume

of knee effusion-synovitis. Second, the MRI method used may have limited the ability to effectively measure synovial inflammation, as the absence of contrast media prevented differentiation between effusion and synovitis. Third, the results of this clinical trial are not generalizable to people without knee effusions. Fourth, no dietary information on omega-3 intake was collected; effects of a higher dietary intake of omega-3 on study outcomes could not be assessed.

Conclusions

Among people with knee OA who have significant knee pain and effusion-synovitis on MRI, 2 g/d of daily krill oil supplementation did not improve knee pain over 24 weeks compared with placebo. These findings do not support krill oil for treating knee pain in this population.

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